A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Entropy and energetics

Why does sodium chloride exist, why is it $CaCl_2$ and not CaCl, and why do some reactions that take in heat still happen by themselves? This topic answers with two tools: Born–Haber and solution cycles, which measure the energy holding ionic solids together, and entropy, which counts the ways energy can be shared out and decides whether a change is feasible at all.

SPEC 12.1 Enthalpy alone doesn't decide whether a reaction happens

WHAT THIS SPEC POINT ASKS

Understand that, because some endothermic reactions happen on their own at room temperature, the enthalpy change can't be the only thing that controls whether a reaction occurs.

Endothermic but spontaneous

If "downhill in enthalpy" were the whole story, no endothermic change could ever happen by itself. Plenty do:

Change	What you see	Why it can happen
Ammonium nitrate dissolving in water (cold packs)	solution gets cold	solid lattice → free ions in solution
$Ba(OH)_2{\cdot}8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2 + 2NH_3(g) + 10H_2O(l)$	gets so cold that water freezes the flask to a wet block	3 moles of solid → 13 moles, including a gas and a liquid
$2CH_3COOH + (NH_4)_2CO_3 \rightarrow 2CH_3COONH_4 + H_2O + CO_2(g)$	fizzes and cools	a gas is made
Ice melting above 0 °C		solid → liquid

What these have in common: the products are **more disordered** than the reactants (more particles, more freedom of movement, gases formed). The extra quantity that decides the direction of change is **entropy** (12.2).

- **Feasible** (spontaneous) means a change can happen without a continuous input of energy from outside. It says nothing about how **fast** (12.11).

MARK-SCHEME WORDING

- "ΔS_{system} must be positive"; "(as the compound does dissolve) ΔS_{total} is > 0"; "$T\Delta S_{system}$ larger than ΔH" also allowed (WCH14 Jun 2021 Q15).
- Endothermic reaction of solids that makes a gas: "moles increase from 4 to 7" **and** "a gas (and liquid products) are produced (from solids)". Give both reasons (WCH14 Jan 2025 Q19).

Often asked as: explain why an endothermic dissolving or reaction is feasible; MCQ on which statement explains an endothermic change.

COMMON MISTAKE

Saying $\Delta S_{surroundings}$ is positive for an endothermic change. Heat is taken **from** the surroundings, so $\Delta S_{surroundings}$ is **negative**; it's the system's entropy that makes up for it.

SPEC 12.2 Entropy: dispersal of particles and energy

WHAT THIS SPEC POINT ASKS

Understand entropy, S, as a measure of the disorder of a system: the random spreading out of molecules and of quanta of energy among them.

What entropy measures

- **Entropy (S)** measures disorder: the **number of ways the particles can be arranged and the energy can be shared out among them**.
- Energy in molecules comes in fixed packets (**quanta**). The more particles there are, and the more freely they move, the more ways there are to share the quanta. The more ways there are, the more likely that state is, and the higher its entropy.
- Units: **J K^{-1} mol^{-1}** (note: joules, not kilojoules).

Counting ways: a feel for it

Share **3 identical quanta** between particles:

Particles	Possible shares	Number of ways
2	(3,0) (2,1) (1,2) (0,3)	4

Particles	Possible shares	Number of ways
3	(3,0,0) (0,3,0) (0,0,3) (2,1,0) (2,0,1) (1,2,0) (0,2,1) (1,0,2) (0,1,2) (1,1,1)	10

Add one particle and the number of ways more than doubles. With real numbers of particles (10^{23}) the number of ways is astronomically large, so a system almost always ends up in the arrangement with the most ways: the most disordered one.

EXAM TIP

Two ideas earn the mark: **disorder** (arrangement of particles) and the **number of ways of distributing energy quanta**. Use either, but tie it to the specific change (e.g. "fewer moles of gas, so fewer ways of distributing energy").

MARK-SCHEME WORDING

- "disorder / number of ways of distributing energy quanta decreases as the number of molecules (of gas) decreases". DO NOT AWARD just "ΔS_{system} is negative" (WCH14 Oct 2020 Q19).
- "a measure of the number of ways that energy can be distributed or particles arranged" (examiner report, Jun 2024).
- "moles (decrease) from 5 to 3", "decrease in disorder" (WCH14 Jun 2024 Q15).

Often asked as: explain the sign of ΔS_{system} in words; MCQ ordering entropies.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Boltzmann's equation $S = k \ln W$ (W = number of ways) is a neat summary but isn't required. The **number-of-ways wording** is rewarded (examiner report, Jun 2024).

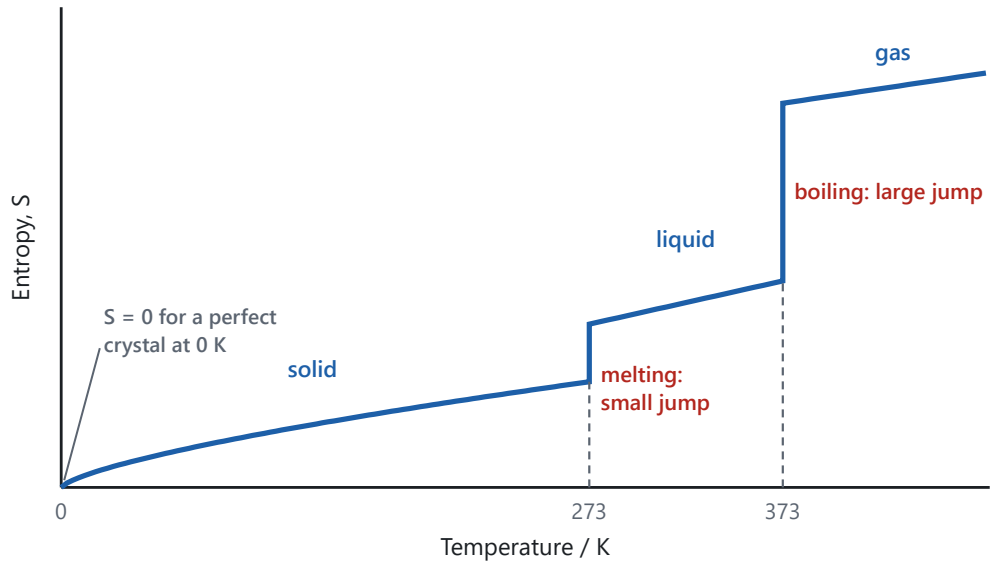
COMMON MISTAKE

Talking only about physical states when the question is about a change in the **number of moles of gas** (examiner report, Jun 2024).

SPEC 12.3 How entropy depends on temperature and state

WHAT THIS SPEC POINT ASKS

Know that a substance's entropy increases with temperature, increases from solid → liquid → gas, and that a perfect crystal at 0 K has zero entropy.



Entropy rises steadily with temperature, jumps a little at melting and a lot at boiling.

- **0 K, perfect crystal: S = 0.** Every particle is fixed in a perfect lattice and there's no energy to share, so there is only **one** possible arrangement.
- **Heating** a substance gives its particles more energy quanta to share, so S rises gradually.
- **Melting**: particles go from fixed positions to moving past each other: a **small jump** in S.
- **Boiling**: particles become completely free and far apart: a **big jump** in S. So S(gas) ≫ S(liquid) > S(solid).

Comparing entropies without data

1. **State** first: gas ≫ liquid > solid.
2. Same state: **more atoms per molecule** (more complex) → more ways to store energy (rotating, vibrating) → higher S. Heavier particles also tend to have higher S.
3. Hard, strongly bonded solids (diamond) have very low S; softer solids higher.

Substance	$S^{\ominus}$ / J K^{-1} mol^{-1}
C(s, diamond)	2.4

Substance	$S^{\ominus}$ / J K^{-1} mol^{-1}
C(s, graphite)	5.7
Mg(s)	32.7
NaCl(s)	72.1
$H_2O(l)$	69.9
$Br_2(l)$	152.2
$H_2O(g)$	188.7
$O_2(g)$	205.0
$CO_2(g)$	213.6
$Br_2(g)$	245.4

MARK-SCHEME WORDING

- Sketch S against T: "general shape of increase from left to right" (a straight line is allowed); "two vertical stages for melting and boiling"; mark 273 K and 373 K for water (WCH14 Jun 2021 Q20).
- Ordering: larger, more complex gas molecules have more entropy than simple gases; a liquid metal such as Hg has less than most gases (WCH14 Jan 2025 Q14).

Often asked as: sketch S against T; MCQ ordering substances by entropy (e.g. Os < Hg < He < O_2).

COMMON MISTAKE

Saying elements have zero entropy, by analogy with $\Delta_f H$. Only a **perfect crystal at 0 K** has S = 0; every substance at 298 K has a positive S.

SPEC 12.4 The natural direction of change: total entropy increases

WHAT THIS SPEC POINT ASKS

Interpret the natural direction of change as the direction in which total entropy increases (ΔS_{total} positive), including gases spreading through a room.

- Left alone, changes go the way that **increases the total entropy** (system + surroundings). That is the second law of thermodynamics in Unit 4 language.

- **Gas spreading**: open a jar of bromine in a room. Bromine molecules spread until they fill the room evenly. They never gather back into the jar, because the spread-out state has vastly more arrangements.
  - One molecule: chance of finding it in the original half of a container = ½.
  - N molecules: chance of finding them all there = $(½)^N$. For N = 100 that is about 10^{-30}; for a mole, effectively zero.
- Heat flows from hot to cold for the same reason: the energy quanta become spread over more particles.
- Feasible means $\Delta S_{total} > 0$. At equilibrium the total entropy of the mixture is at its maximum and nothing further changes overall (13.8).

MARK-SCHEME WORDING

- "chemical reactions are feasible when ΔS_{total} is positive" (WCH14 Jan 2026 Q17).

Often asked as: the opening point of a feasibility answer; rarely asked on its own.

COMMON MISTAKE

Saying a reaction happens because "the system's entropy increases". The test is the **total** entropy.

SPEC 12.5 Entropy changes: changes of state, dissolving and changing numbers of moles

WHAT THIS SPEC POINT ASKS

Explain why entropy changes during (i) changes of state, (ii) dissolving a solid ionic lattice, (iii) reactions where the number of moles changes from reactants to products.

(i) Changes of state

Change	ΔS_{system}	Why
melting, boiling, subliming	**positive**	particles gain freedom; boiling gives the biggest increase
freezing, condensing	**negative**	particles become more ordered

(ii) Dissolving an ionic solid

Two effects pull in opposite directions:

- The **lattice breaks up**: ions go from fixed positions to moving freely in solution → entropy **increases**.
- Water molecules become **ordered round each ion** (hydration shells) → entropy **decreases**. This effect is biggest for **small, highly charged ions** (Mg^{2+}, Al^{3+}), which hold water tightly.

So ΔS_{system} for dissolving is usually positive (e.g. about +43 J K^{-1} mol^{-1} for NaCl, +109 for NH_4NO_3) but can be negative (about −115 for $MgCl_2$). Don't assume the sign.

(iii) Changes in the number of moles

- **More moles of gas** in the products → ΔS_{system} **positive** and large.
- Fewer moles of gas → negative. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$: 4 moles of gas → 2, so ΔS_{system} is negative (−199 J K^{-1} mol^{-1}).
- With no gases, count all moles, and look at state changes: $Ba(OH)_2 \cdot 8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2(s) + 2NH_3(g) + 10H_2O(l)$ goes from 3 moles of solid to 13 moles, including a gas.

EXAM TIP

Predict the sign in this order: (1) any change in moles of **gas**? That dominates. (2) Any solid → liquid or solution? (3) Then total moles.

MARK-SCHEME WORDING

- "1 mole of reactant becomes 7 moles of products (and more moles have higher entropy)" OR "a solid becomes (a solid and) a liquid (and liquids have higher entropy than solids)" (WCH14 Oct 2025 Q9). "mols / molecules / particles" are all allowed.
- "one molecule (of reactant) forms two molecules (of product) in reaction 1"; "no change in the number of molecules in reaction 2" (WCH14 Oct 2022 Q17).

Often asked as: predict and explain the sign of ΔS_{system} with no data; explain why solids don't react but their solutions do.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Why a mixture of two ionic solids doesn't react but their solutions do: "in solids the particles are fixed in a lattice (so unable to move)", "so they cannot collide and react", "but when dissolved the particles are mobile" (WCH14 Jan 2025 Q19). This is AS collision theory applied here.

COMMON MISTAKE

Vague "it's more disordered" answers. **Count** the molecules or name the change of state.

SPEC 12.6 Total entropy change = system + surroundings

WHAT THIS SPEC POINT ASKS

Know that the total entropy change of a reaction is the sum of the entropy change of the system and that of the surroundings: $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$.

Entropy change of the surroundings and total entropy change

$$\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T} \qquad \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

ΔH in J mol^{-1} (× 1000 from kJ) T in K Entropies in J K^{-1} mol^{-1}

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$$

- **System** = the reacting chemicals. ΔS_{system} comes from the entropies of products and reactants (12.7).
- **Surroundings** = everything else (the solvent, flask, air). $\Delta S_{surroundings}$ comes from the heat the reaction gives out or takes in (12.8).
- Only **ΔS_{total}** decides feasibility: positive means feasible.

Exothermic or endothermic?	$\Delta S_{surroundings}$
Exothermic ($\Delta H < 0$): heat goes **into** the surroundings	**positive**
Endothermic ($\Delta H > 0$): heat comes **out of** the surroundings	**negative**

MARK-SCHEME WORDING

- "$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$ and $\Delta S_{surroundings} = -\Delta H \div T$" (WCH14 Jan 2026 Q17).

Often asked as: the first indicative point in a feasibility 6-mark answer.

COMMON MISTAKE

Adding ΔS_{system} in J K^{-1} mol^{-1} to $\Delta S_{surroundings}$ in kJ K^{-1} mol^{-1}. Put both in **J K^{-1} mol^{-1}** before adding.

SPEC 12.7 Calculating ΔS_{system}

WHAT THIS SPEC POINT ASKS

Calculate the entropy change of the system for a reaction from the entropies of the reactants and products.

Entropy change of the system

$$\Delta S^\circ_{\text{system}} = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$$

Multiply each standard entropy by its number of moles in the equation Units: J K⁻¹ mol⁻¹

$$\Delta S_{system} = \Sigma S(products) - \Sigma S(reactants)$$

- **Multiply** each S value by its number of moles in the equation.
- Elements have **non-zero** entropies: include them.
- Use the value for the right **state** ($H_2O(l)$ 69.9, $H_2O(g)$ 188.7).
- Units J K⁻¹ mol⁻¹, with a **sign**.

WORKED EXAMPLE

Worked example: three reactions

Haber process: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

S: N_2 191.6, H_2 130.6, NH_3 192.3 J K⁻¹ mol⁻¹

$\Delta S_{system} = (2 \times 192.3) - (191.6 + 3 \times 130.6) = 384.6 - 583.4 =$ **−198.8 J K⁻¹ mol⁻¹** (4 mol gas → 2 mol gas)

Decomposing limestone: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

S: $CaCO_3$ 92.9, CaO 39.7, CO_2 213.6

$\Delta S_{system} = (39.7 + 213.6) - 92.9 =$ **+160.4 J K⁻¹ mol⁻¹** (a gas is made)

Burning magnesium: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

S: Mg 32.7, O_2 205.0, MgO 26.9

$\Delta S_{system} = (2 \times 26.9) - (2 \times 32.7 + 205.0) = 53.8 - 270.4 =$ **−216.6 J K⁻¹ mol⁻¹** (a gas is used up)

ΔS for $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

1. Data / J K^{-1} mol^{-1}: $S^{\ominus}[N_2(g)]$ = 191.6; $S^{\ominus}[H_2(g)]$ = 130.6; $S^{\ominus}[NH_3(g)]$ = 192.3.
2. Products: 2 × 192.3 = 384.6. Reactants: 191.6 + 3 × 130.6 = 583.4.
3. $$\Delta S_{\text{system}} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$$

Negative, as predicted: four moles of gas become two.

MARK-SCHEME WORDING

- "Σ entropy of products − Σ entropy of reactants" with values substituted, e.g. (109.2 + 6 × 69.9) − 343 = +185.6 J K^{-1} mol^{-1} (WCH14 Oct 2025 Q9).
- Sign **and** units J K^{-1} mol^{-1}. "Penalise incorrect units once only."

Often asked as: in almost every paper, as the first part of a ΔS_{total} or feasibility-temperature calculation.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Working backwards to the **S value of one substance** from a given ΔS_{system} (WCH14 Jan 2022 Q22).
- Finding ΔS_{system} from $\Delta S_{total} - \Delta S_{surroundings}$ (WCH14 Oct 2020 Q19).

COMMON MISTAKE

Forgetting to multiply by 2 or 3. Subtracting products from reactants. Quoting 4 or more SF when the data are to 3 or 4 SF.

SPEC 12.8 $\Delta S_{surroundings} = -\Delta H/T$, and ΔS_{total}

WHAT THIS SPEC POINT ASKS

Calculate the entropy change of the surroundings using $\Delta S_{surroundings} = -\Delta H/T$, and hence ΔS_{total}.

$$\Delta S_{surroundings} = -\Delta H \div T$$

- ΔH must be in **J mol^{-1}** (multiply kJ by 1000). T in **kelvin** (°C + 273).
- The **minus sign**: an exothermic reaction (ΔH negative) gives heat to the surroundings, so their entropy **rises**.

- Dividing by **T**: the same amount of heat makes a **bigger difference to cold surroundings** than to hot ones, which are already highly disordered. (Like a shout in a quiet library compared with a busy street.)

WORKED EXAMPLE

Worked example: ΔS_{total} for the Haber process at 298 K

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; $\Delta H = -92.2$ kJ mol^{-1}; $\Delta S_{system} = -198.8$ J K^{-1} mol^{-1}

$\Delta S_{surroundings} = -\Delta H \div T = -(-92\,200) \div 298 =$ **+309.4 J K^{-1} mol^{-1}**

$\Delta S_{total} = -198.8 + 309.4 =$ **+110.6 J K^{-1} mol^{-1}**

Positive, so the reaction is **feasible** at 298 K (although it is extremely slow without a catalyst).

WORKED EXAMPLE

Worked example: ΔS_{total} for burning magnesium

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$; $\Delta H = -1203.4$ kJ mol^{-1}; $\Delta S_{system} = -216.6$ J K^{-1} mol^{-1}

$\Delta S_{surroundings} = +1\,203\,400 \div 298 = +4038$ J K^{-1} mol^{-1}

$\Delta S_{total} = -216.6 + 4038 =$ **+3822 J K^{-1} mol^{-1}**: hugely positive, so feasible.

WORKED EXAMPLE

Worked example: back to ΔH

An endothermic reaction at 298 K has $\Delta S_{surroundings} = -234.9$ J K^{-1} mol^{-1}. Find ΔH.

$\Delta H = -T \times \Delta S_{surroundings} = -298 \times (-234.9) = +70\,000$ J mol^{-1} = **+70.0 kJ mol^{-1}**

MARK-SCHEME WORDING

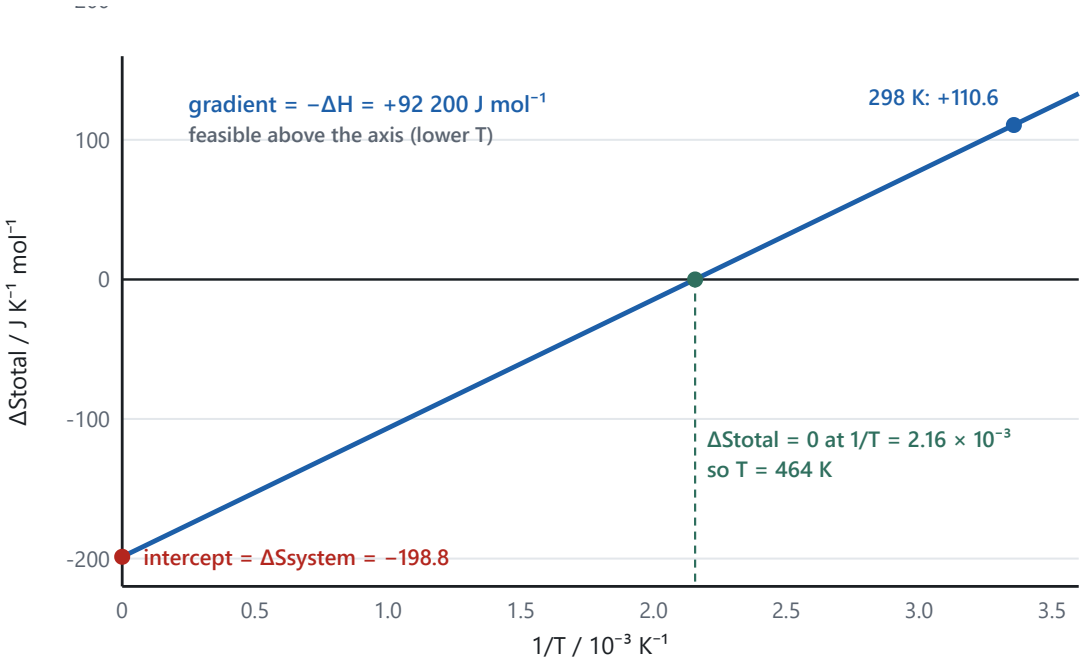
- "$= -(-631.3 \times 10^3) \div 298 = +2118.5$ J K^{-1} mol^{-1}" (WCH14 Jan 2020 Q19): the kJ → J conversion and K are both checked.
- Temperature in °C must be converted, e.g. 23.0 + 273 = 296 K (WCH14 Jun 2024 Q15).
- Sign **and** units required, e.g. −295.64 J K^{-1} mol^{-1} or −0.29564 kJ K^{-1} mol^{-1} (WCH14 Oct 2025 Q9).

Often asked as: calculate $\Delta S_{surroundings}$ and ΔS_{total} and comment on feasibility; work backwards to ΔH.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **ΔS_{total} against 1/T graph.** $\Delta S_{total} = \Delta S_{system} - \Delta H \times (1/T)$, a straight line: **gradient = −ΔH**, **intercept = ΔS_{system}**, and where the line crosses $\Delta S_{total} = 0$ gives the temperature at which feasibility changes (WCH14 Jan 2022 Q22).
- Back-calculating $\Delta H = -\Delta S_{surroundings} \times T$ (WCH14 Jan 2025 Q19).



$N_2 + 3H_2 \rightarrow 2NH_3$: gradient = $-\Delta H$ = +92 200 J mol^{-1}, intercept = ΔS_{system} = −198.8 J K^{-1} mol^{-1}. ΔS_{total} is positive (feasible) below 464 K, i.e. to the right of 1/T = 2.16 × 10^{-3} K^{-1}.

COMMON MISTAKE

The top three: forgetting ×1000; dropping the minus sign (especially when ΔH is already negative: −(−) = +); using °C.

SPEC 12.9 Feasibility: balancing the entropy terms, and the effect of temperature

WHAT THIS SPEC POINT ASKS

Understand that whether a reaction is feasible depends on (i) the balance between ΔS_{system} and $\Delta S_{surroundings}$, so even endothermic reactions can be spontaneous, and (ii) temperature, because a higher temperature makes $\Delta S_{surroundings}$ smaller in magnitude. Calculate the temperature at which a reaction becomes feasible. ($\Delta G = \Delta H - T\Delta S_{system}$ may be used but isn't required.)

The four combinations

ΔS_{system} hardly changes with temperature (unless something changes state), and neither does ΔH. Only the $\Delta S_{surroundings}$ term, $-\Delta H/T$, changes, and **it gets smaller in size as T rises**.

ΔH	ΔS_{system}	$\Delta S_{surroundings}$	Feasible?
negative (exo)	positive	positive	**always**: both terms positive
negative (exo)	negative	positive	at **low T**, while $-\Delta H/T$ is bigger than the size of ΔS_{system}
positive (endo)	positive	negative	at **high T**, once ΔS_{system} outweighs $\Delta H/T$
positive (endo)	negative	negative	**never**: both terms negative

Temperature at which feasibility changes

Set $\Delta S_{total} = 0$: $\Delta S_{system} - \Delta H/T = 0$, so $\mathbf{T = \Delta H \div \Delta S_{system}}$ (ΔH in J mol^{-1}).

WORKED EXAMPLE

Worked example: decomposing calcium carbonate

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$; $\Delta H = +178$ kJ mol^{-1}; $\Delta S_{system} = +160.4$ J K^{-1} mol^{-1}

At 298 K: $\Delta S_{surroundings} = -178\,000 \div 298 = -597.3$; $\Delta S_{total} = 160.4 - 597.3 = \mathbf{-436.9\ J\ K^{-1}\ mol^{-1}}$: not feasible.

Feasible when $\Delta S_{total} \geq 0$: T = 178 000 ÷ 160.4 = **1110 K** (about 837 °C). Above this, $CaCO_3$ decomposes: this is why a lime kiln runs at over 900 °C.

WORKED EXAMPLE

Worked example: the Haber process

$\Delta H = -92.2$ kJ mol^{-1}, $\Delta S_{system} = -198.8$ J K^{-1} mol^{-1}: exothermic with a negative ΔS_{system}.

T = −92 200 ÷ −198.8 = **464 K**. The reaction is feasible **below** 464 K. The industrial plant runs at about 700 K for **rate**, so ΔS_{total} there is negative and the equilibrium yield is low: that is the compromise (13.4, 13.8).

Using ΔG instead (allowed, not required)

ΔG = ΔH − TΔS_{system}, and ΔG = −TΔS_{total}. A reaction is feasible when **ΔG is negative** (ΔS_{total} positive). Setting ΔG = 0 gives the same T = ΔH/ΔS_{system}.

Is decomposing calcium carbonate feasible at 298 K?

1. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. Data: $\Delta_f H$ / kJ mol^{-1}: $CaCO_3$ −1206.9, CaO −635.1, CO_2 −393.5. $S^{\ominus}$ / J K^{-1} mol^{-1}: $CaCO_3$ 92.9, CaO 39.7, CO_2 213.6.

2. Enthalpy change: ΔH = (−635.1 − 393.5) − (−1206.9) = +178.3 kJ mol^{-1}.

3. Entropy change of the system: ΔS_{system} = (39.7 + 213.6) − 92.9 = +160.4 J K^{-1} mol^{-1}.

   Positive, because a gas is made from a solid.

4. Entropy change of the surroundings at 298 K, converting ΔH to J mol^{-1}:

$$\Delta S_{\text{surroundings}} = -\frac{178\,300}{298} = -598.3\ \text{J K}^{-1}\,\text{mol}^{-1}$$

5. Total:

$$\Delta S_{\text{total}} = +160.4 + (-598.3) = -437.9\ \text{J K}^{-1}\,\text{mol}^{-1}$$

   Negative, so limestone does not decompose at room temperature.

6. At what temperature does it become feasible? At the boundary ΔS_{total} = 0, so ΔS_{system} = ΔH/T:

$$T = \frac{\Delta H}{\Delta S_{\text{system}}} = \frac{178\,300}{160.4} = 1110\ \text{K}$$

   Above about 1110 K (840 °C), Δ$S_{surroundings}$ is small enough for the positive ΔS_{system} to win. Lime kilns run hotter than this.

From ΔS_total to ΔG

1. Start from the total entropy change:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} - \frac{\Delta H}{T}$$

2. Multiply every term by −T:

$$-T\Delta S_{\text{total}} = \Delta H - T\Delta S_{\text{system}}$$

3. Define the right-hand side as the Gibbs energy change:

$$\Delta G = \Delta H - T\Delta S_{\text{system}} = -T\Delta S_{\text{total}}$$

Because T is always positive, ΔG is negative exactly when ΔS_{total} is positive.

Gibbs energy change

$$\Delta G = \Delta H - T\Delta S_{\text{system}}$$

ΔG, ΔH in kJ mol^{-1} ΔS in kJ K^{-1} mol^{-1} (÷ 1000 from J) T in K

ΔG for decomposing calcium carbonate

1. ΔH = +178.3 kJ mol^{-1} and ΔS_{system} = +160.4 J K^{-1} mol^{-1} = +0.1604 kJ K^{-1} mol^{-1}.

2. At 298 K:

$$\Delta G = 178.3 - 298 \times 0.1604 = +130.5\ \text{kJ mol}^{-1}$$

Positive: not feasible, as the entropy calculation showed.

3. At 1200 K:

$$\Delta G = 178.3 - 1200 \times 0.1604 = -14.2\ \text{kJ mol}^{-1}$$

Negative: feasible at kiln temperatures.

MARK-SCHEME WORDING

- Feasibility 6-mark (WCH14 Jan 2026 Q17): exothermic + positive ΔS_{system}: always feasible; exothermic + negative ΔS_{system}: feasible only if $-\Delta H/T > |\Delta S_{system}|$; endothermic + positive ΔS_{system}: feasible only if $\Delta S_{system} > \Delta H/T$; endothermic + negative: never feasible. The ΔG route is allowed throughout.

- Always feasible: "$\Delta S_{surroundings}$ and ΔS_{system} are (both) positive so ΔS_{total} will always be positive". Answers mentioning only ΔS_{total} or only $\Delta S_{surroundings}$ scored zero (WCH14 Jun 2022).
- Higher T: "$\Delta S_{surroundings}$ will decrease (become less positive); ΔS_{system} does not change (significantly); so ΔS_{total} decreases".
- $T = \Delta H \div \Delta S_{system}$ = 88 100 ÷ 185.6 = 474.68 K (WCH14 Oct 2025 Q9).

Often asked as: minimum temperature for feasibility (most papers); why feasible at all temperatures; why a higher temperature lowers ΔS_{total}; 6-mark on the four combinations; Haber temperature as a compromise.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- "Justify" a plant temperature means using the entropy/yield data, not reciting collision theory (examiner report, Jan 2020).
- A feasibility temperature that's impractically high: "the temperature required is too high" (WCH14 Oct 2023 Q16).
- If ΔH is close to zero, $\Delta S_{surroundings}$ hardly changes with T, so feasibility (and K) barely change (WCH14 Jan 2024 Q19).

COMMON MISTAKE

Writing "$\Delta S_{surroundings}$ gets bigger/smaller" without saying "more positive" or "less negative". Be precise about signs.

Bullet lists in the 6-mark question: link the points in sentences.

Mixing J and kJ in $T = \Delta H \div \Delta S_{system}$.

SPEC 12.10 A reaction can go even if one entropy term is negative

WHAT THIS SPEC POINT ASKS

Understand that a reaction can happen as long as ΔS_{total} is positive, even if ΔS_{system} or $\Delta S_{surroundings}$ is negative.

- What matters is the **sum**. Compare the **sizes** of the two terms.
- **ΔS_{system} negative, $\Delta S_{surroundings}$ positive and bigger**: burning magnesium (−216.6 + 4038 = +3822); the Haber process at 298 K (−198.8 + 309.4 = +110.6); freezing water below 0 °C.
- **$\Delta S_{surroundings}$ negative, ΔS_{system} positive and bigger**: dissolving ammonium nitrate at 298 K:

WORKED EXAMPLE

Worked example: an endothermic change that happens

$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$; $\Delta_{sol}H = +25.7$ kJ mol^{-1}; $\Delta S_{system} = +108.7$ J K^{-1} mol^{-1}

$\Delta S_{surroundings} = -25\,700 \div 298 = -86.2$ J K^{-1} mol^{-1}

$\Delta S_{total} = 108.7 - 86.2 =$ **+22.5 J K^{-1} mol^{-1}** → feasible, so ammonium nitrate dissolves even though the water gets cold.

MARK-SCHEME WORDING

- The verdict is "ΔS_{total} is positive (so the reaction is feasible)", backed by the numbers (WCH14 Jan 2020 Q19: ΔS_{system} −337.8, $\Delta S_{surroundings}$ +2118.5).

Often asked as: part of 12.8/12.9 calculations, ending "comment on your answer".

COMMON MISTAKE

Concluding "not feasible" because ΔS_{system} is negative. Always finish with ΔS_{total}.

Forgetting that "feasible" doesn't mean "fast": if asked why a feasible reaction doesn't happen, talk about activation energy (12.11).

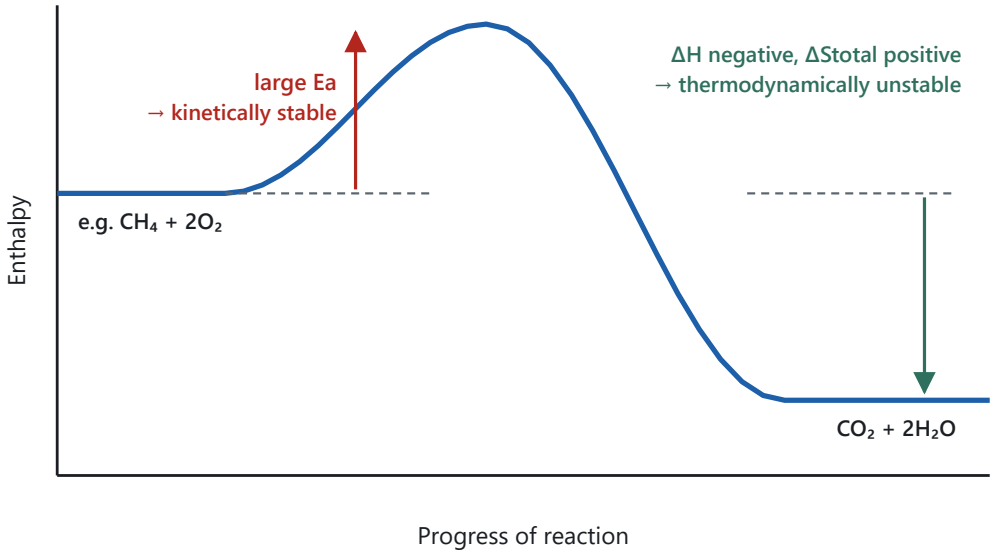
SPEC 12.11 Thermodynamic stability and kinetic stability

WHAT THIS SPEC POINT ASKS

Understand and tell apart thermodynamic stability and kinetic stability.

	Thermodynamic stability	Kinetic stability
Question it answers	**Can** it react? (Which way is "downhill" overall?)	**How fast** will it react?
Decided by	**ΔS_{total}** (or ΔG, or K)	**activation energy, E_a** (and so the rate)
Stable means	ΔS_{total} for the reaction is **negative**: the reaction isn't feasible	E_a is **so high** that the rate is negligible at that temperature

- A substance can be **thermodynamically unstable but kinetically stable**: the reaction is feasible ($\Delta S_{total} > 0$) but too slow to notice because E_a is high.
  - **Methane in air**: combustion is extremely feasible, but nothing happens until a spark supplies the activation energy.
  - **Diamond → graphite** is feasible at room temperature, but the strong C–C bonds give a huge E_a, so diamonds last.
  - **Hydrogen peroxide** decomposes slowly; MnO_2 (a catalyst) lowers E_a and it fizzes.
  - **SiO_2** in a feasible reaction at room temperature: no reaction, because breaking strong Si–O bonds in a giant lattice needs a lot of energy.
- **Thermodynamically stable** means it won't react however long you wait (e.g. $CaCO_3$ decomposing at 298 K: $\Delta S_{total} = -437$ J K^{-1} mol^{-1}).



Feasible (products far below reactants, ΔS_{total} positive) but kinetically stable (large E_a): the reaction needs a spark, heat or a catalyst to start.

MARK-SCHEME WORDING

- "(thermodynamically) feasible because ΔS_{total} is positive"; "activation energy (is) high so the reaction is very slow (at low temperatures)" (WCH14 Jun 2023 Q20).
- "a large amount of energy is needed to break the (strong Si–O) bonds" (WCH14 Jun 2024 Q15). "Kinetically stable" on its own is ignored: give the reason.
- "a lot of energy is required to break the strong C–C bonds in cyclopropane" (WCH14 Jan 2020 Q18).
- Only call a reaction "kinetically stable / too slow" if your ΔS_{total} is **positive**. If ΔS_{total} is negative the reaction simply isn't feasible (WCH14 Oct 2025 Q9).

Often asked as: why a feasible reaction doesn't happen at room temperature; MCQ "kinetically stable and thermodynamically unstable".

COMMON MISTAKE

Explaining "no reaction" with collision frequency of solids, or with intermolecular forces. The point is the **energy needed to break strong bonds** (E_a) (examiner report, Jun 2024).

Commenting only on ΔS_{total} when the question also wants the rate (examiner report, Jun 2023).

SPEC 12.12 Definitions: atomisation, electron affinity, lattice energy

WHAT THIS SPEC POINT ASKS

Define (i) the standard enthalpy change of atomisation, $\Delta_{at}H$; (ii) electron affinity; (iii) lattice energy, as the exothermic change when one mole of an ionic solid forms from its gaseous ions.

The definitions, each with an equation

Term	Definition	Example equation	Sign
Standard enthalpy change of atomisation, $\Delta_{at}H^{\ominus}$	enthalpy change when **one mole of gaseous atoms** is formed from the element in its **standard state**, under standard conditions	$Na(s) \rightarrow Na(g)$, +107 kJ mol^{-1} $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$, +122 kJ mol^{-1} $\frac{1}{2}Br_2(l) \rightarrow Br(g)$ $\frac{1}{2}I_2(s) \rightarrow I(g)$	always **positive**
First electron affinity	enthalpy change when **each atom in one mole of gaseous atoms gains one electron**, forming one mole of gaseous 1− ions	$Cl(g) + e^- \rightarrow Cl^-(g)$, −349 kJ mol^{-1} $O(g) + e^- \rightarrow O^-(g)$, −141 kJ mol^{-1}	usually **negative**
Second electron affinity	enthalpy change when each ion in one mole of gaseous 1− ions gains one electron, forming one mole of gaseous 2− ions	$O^-(g) + e^- \rightarrow O^{2-}(g)$, +798 kJ mol^{-1}	**positive**
Lattice energy (LE)	enthalpy change when **one mole of an ionic solid is formed from its gaseous ions** (under standard conditions)	$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$, −787 kJ mol^{-1} $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$	always **negative**

You also need, from Units 1 and 2: **first ionisation energy** ($Na(g) \rightarrow Na^+(g) + e^-$, always positive), **second ionisation energy** ($Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$) and **enthalpy change of formation** ($Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$).

Why the signs are what they are

- **Atomisation**: bonds (metallic or covalent) are broken: endothermic.
- **First EA**: the incoming electron is attracted by the nucleus: energy released.
- **Second EA**: the electron is added to an ion that is **already negative**, so it is **repelled**. Energy must be put in: endothermic.
- **Lattice energy**: oppositely charged ions attract and come together: energy released. The more negative it is, the stronger the ionic bonding. It can't be measured directly, so it's found from a Born–Haber cycle (12.13).

MARK-SCHEME WORDING

- Atomisation MCQ: "one mole of gaseous atoms", and "the standard state of iodine is solid", so $\frac{1}{2}I_2(s) \rightarrow I(g)$ (WCH14 Jan 2025 Q6); bromine: $\frac{1}{2}Br_2(l) \rightarrow Br(g)$ (WCH14 Jan 2021).
- Second EA: $O^-(g) + e^- \rightarrow O^{2-}(g)$. The first and second together are not one step (WCH14 Jan 2025 Q7).
- Why second EA is endothermic: "the electron is being added to a negative ion and so there is repulsion (so energy is required)" (WCH14 Jun 2022 Q21).

Often asked as: pick the equation that represents a defined quantity (MCQ); explain the sign of the second electron affinity.

COMMON MISTAKE

Writing $Cl_2(g) \rightarrow 2Cl(g)$ for the atomisation of chlorine: that's two moles of atoms. Use $\frac{1}{2}Cl_2$.

Saying an electron affinity is "larger" or "higher" when you mean more or less exothermic.

Confusing electron affinity (gaining electrons) with ionisation energy (losing them).

SPEC 12.13 Born–Haber cycles

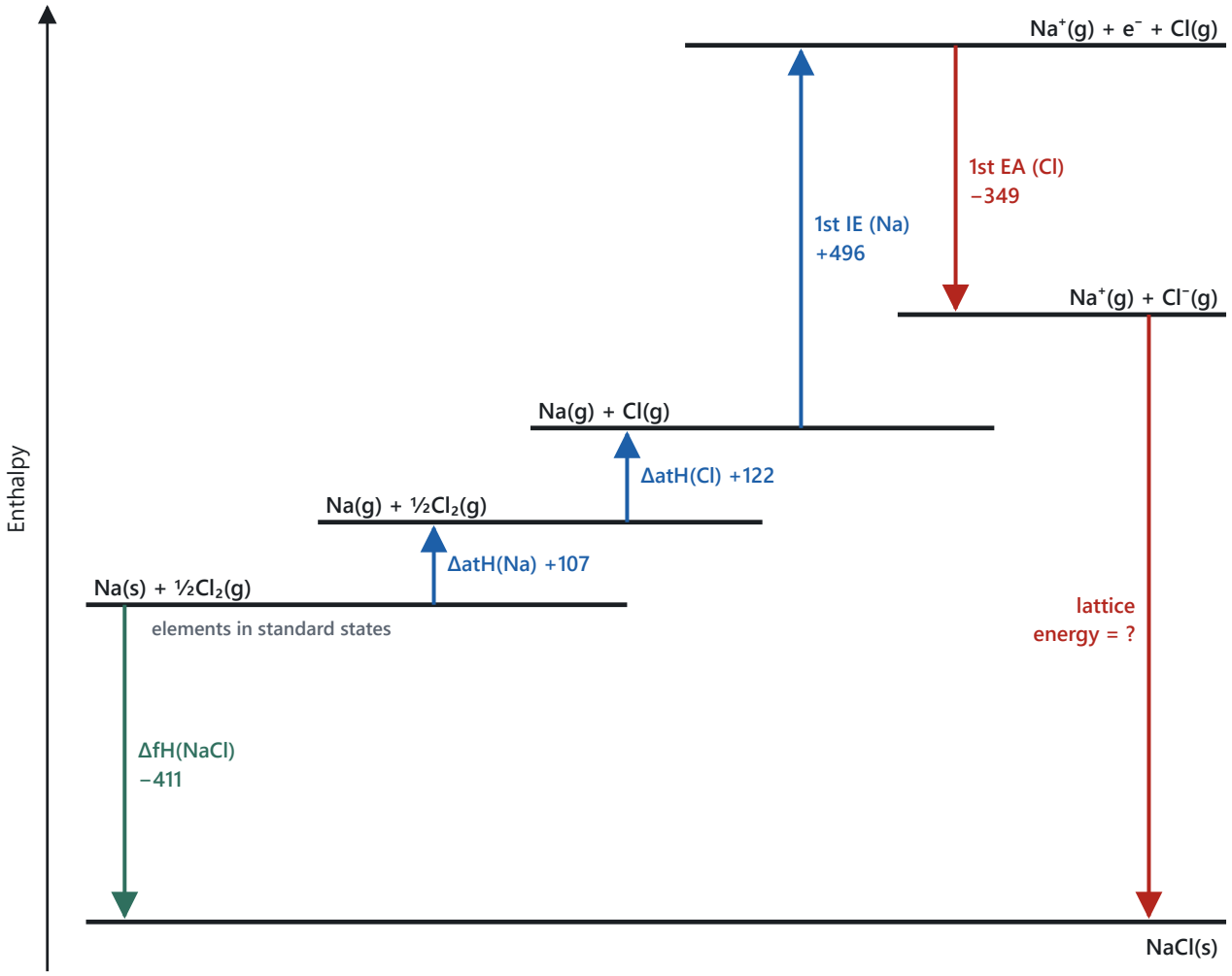
WHAT THIS SPEC POINT ASKS

Construct Born–Haber cycles and use them to calculate an unknown enthalpy change.

Building the cycle as an energy-level diagram

1. Draw a line for the **elements in their standard states**, about a third of the way up. Energy increases up the page.

2. **Upward** (endothermic) steps: atomise the metal, atomise the non-metal, ionise the metal (first, then second, ionisation energy).
3. **Downward** steps for exothermic first electron affinities; an **upward** step for a second electron affinity.
4. You now have **gaseous ions**. The **lattice energy** arrow goes **down** from here to the ionic solid.
5. The **enthalpy of formation** links the elements line to the ionic solid (usually down).
6. Every line has species with **state symbols**, and **one** arrowhead per arrow.



LE = −411 − (107 + 122 + 496 − 349) = −787 kJ mol^{-1}

Born–Haber cycle for NaCl. Going round either way from the elements to NaCl(s) gives the same total (Hess's law).

The calculation

By Hess's law, the direct route ($\Delta_f H$) equals the long way round:

$\Delta_f H$ = $\Delta_{at} H$(metal) + $\Delta_{at} H$(non-metal) + ionisation energies + electron affinities + LE

Rearrange for whichever value is missing.

Born–Haber cycle for a compound MX

$$\Delta_{\mathrm{f}}H = \Delta_{\mathrm{at}}H(\mathrm{M}) + \mathrm{IE}(\mathrm{M}) + \Delta_{\mathrm{at}}H(\mathrm{X}) + \mathrm{EA}(\mathrm{X}) + \mathrm{LE}(\mathrm{MX})$$

IE: ionisation energy (or energies) EA: electron affinity (or affinities) LE: lattice energy

WORKED EXAMPLE

Worked example 1: lattice energy of NaCl

$\Delta_fH(NaCl) = -411$; $\Delta_{at}H(Na) = +107$; $IE_1(Na) = +496$; $\Delta_{at}H(Cl) = +122$; $EA_1(Cl) = -349$ (all kJ mol^{-1})

$LE = \Delta_fH - (\Delta_{at}H(Na) + IE_1 + \Delta_{at}H(Cl) + EA_1)$

$= -411 - (107 + 496 + 122 - 349) = -411 - 376 =$ **−787 kJ mol^{-1}**

WORKED EXAMPLE

Worked example 2: lattice energy of $MgCl_2$ (a 2:1 compound)

$\Delta_fH(MgCl_2) = -641$; $\Delta_{at}H(Mg) = +148$; $IE_1 + IE_2$ of Mg $= +738 + 1451$; $\Delta_{at}H(Cl) = +122$; $EA_1(Cl) = -349$

Two chloride ions, so **double** the chlorine atomisation and electron affinity:

Sum of steps to $Mg^{2+}(g) + 2Cl^-(g) = 148 + 738 + 1451 + 2(122) + 2(-349) = +1883$

$LE = -641 - 1883 =$ **−2524 kJ mol^{-1}**

WORKED EXAMPLE

Worked example 3: second electron affinity of oxygen, from MgO

$\Delta_fH(MgO) = -602$; $\Delta_{at}H(Mg) = +148$; $IE_1 + IE_2 = +738 + 1451$; $\Delta_{at}H(O) = +249$; $EA_1(O) = -141$; $LE(MgO) = -3845$

$\Delta_fH = 148 + 738 + 1451 + 249 + (-141) + EA_2 + (-3845)$

$-602 = -1400 + EA_2$

$EA_2 =$ **+798 kJ mol^{-1}**. Positive, as expected: the electron is repelled by O^-.

Lattice energy of sodium chloride

1. Data / kJ mol^{-1}: $\Delta_f H$(NaCl) = −411; $\Delta_{at} H$(Na) = +107; first IE(Na) = +496; $\Delta_{at} H$(Cl) = +122; first EA(Cl) = −349.

2. Go round the indirect route from the elements to the gaseous ions: atomise sodium, ionise it, atomise chlorine, add the electron to chlorine.

$$107 + 496 + 122 + (-349) = +376 \text{ kJ mol}^{-1}$$

So $Na^+(g)$ + $Cl^-(g)$ sits 376 kJ mol^{-1} above the elements.

3. Hess's law: $\Delta_f H$ = 376 + LE, so

$$\text{LE} = \Delta_f H - 376 = -411 - 376 = -787 \text{ kJ mol}^{-1}$$

MARK-SCHEME WORDING

- Cycle marks: "arrows upwards for first and second ionisation energies … downward arrows for electron affinity and lattice enthalpy … correct species including state symbols on horizontal lines" (WCH14 Jan 2020 Q19).
- The unlabelled arrow from elements to compound is "enthalpy (change) of formation" (WCH14 Jan 2026 Q14).
- Calculations: correct answer with working gets full marks; one slip usually costs 1 mark.

Often asked as: complete a cycle with species and arrows; calculate LE, $\Delta_f H$, an electron affinity or a second ionisation energy from a cycle.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Why $MgCl_3$ doesn't exist**: the **third ionisation energy** of Mg (removing an electron from an inner shell) is enormous and "would not be compensated for by the lattice energy", so $\Delta_f H$ would be very positive (WCH14 Jan 2024 Q17).
- Any value in the cycle can be the unknown, including IE_2 of a Group 2 metal (WCH14 Jun 2022 Q21) or $\Delta_f H$ (WCH14 Jan 2026 Q14: −552 kJ mol^{-1}, which must be negative).

COMMON MISTAKE

Not doubling the atomisation and electron affinity of chlorine for $MgCl_2$ (the commonest error in several examiner reports).

Arrowheads on both ends; missing state symbols; arrows in the wrong direction.

A positive $\Delta_f H$ for a stable ionic compound: recheck your signs.

SPEC 12.14 Experimental and theoretical lattice energies: evidence for covalency

WHAT THIS SPEC POINT ASKS

Understand that comparing the experimental lattice energy (from a Born–Haber cycle) with the theoretical value (from electrostatic calculations) shows how much covalent bonding a compound has.

Two ways to get a lattice energy

- **Experimental**: from a Born–Haber cycle, using measured enthalpy changes. This is the **real** value.
- **Theoretical**: calculated from a **100% ionic model**: the ions are **perfect spheres** (point charges) with **no distortion**, touching, held only by electrostatic attraction.

What the comparison tells you

- **Values close** → the ionic model is good → the compound is **almost purely ionic**.
- **Experimental more exothermic** than theoretical → there is **extra** bonding the model leaves out: **partial covalent character** caused by polarisation (12.15). The bigger the gap, the more covalent.

Compound	Experimental LE / kJ mol^{-1}	Theoretical LE / kJ mol^{-1}	Difference	Verdict
NaCl	−780	−770	1%	almost purely ionic
AgCl	−905	−833	9%	some covalency
AgBr	−891	−816	9%	some covalency
AgI	−889	−778	14%	most covalent

(Typical data-book values; use whatever the question gives you.) For MgO, the two values in one paper agreed to within 4 kJ mol^{-1} out of about 3800: O^{2-} is small and hard to polarise, so MgO is essentially ionic (WCH14 Oct 2025 Q3).

MARK-SCHEME WORDING

- Model assumptions: "the bonding is 100% ionic / the ions are in contact / perfect spheres / point charges" (any two) (WCH14 Oct 2024 Q20).
- "the experimental value is more exothermic … the chloride ion is polarised (by the magnesium ion) … some covalent character".
- "calcium chloride is almost completely ionic"; "calcium iodide has partially covalent character"; "the iodide ion is larger, so more (easily) polarised" (WCH14 Jun 2023 Q17).
- MgO v MgS: MgO "very nearly purely ionic because the oxide ion is small and very difficult to polarise"; MgS "the sulfide ion is large and more easily polarised" (WCH14 Jan 2026 Q14). Talk about **ions**, not atoms.
- Use "**more exothermic / less exothermic**", not "bigger / higher" for negative numbers.

Often asked as: 6-mark comparison of lattice energies (WCH14 Jun 2021 Q18: NaCl and MgI_2; Jan 2024 Q17: NaF, MgF_2, $MgCl_2$); 4-mark comparing two calcium or magnesium halides.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Near-identical experimental and theoretical values are **evidence of 100% ionic bonding**, not experimental error (WCH14 Oct 2025 Q3; Jan 2020 examiner report).

COMMON MISTAKE

Blaming the difference on "experimental error".

Writing about atoms, atomic radius, molecules, or "the compound being polarised". It's the **anion** that is polarised, by the **cation**.

Confusing **polarising power** (a cation property) with **polarisability** (an anion property).

SPEC 12.15 Polarisation: how cations make ionic bonds partly covalent

WHAT THIS SPEC POINT ASKS

Understand that polarisation of anions by cations gives some covalent character to an ionic bond, based on evidence from Born–Haber cycles.

What happens

- A cation attracts the electron cloud of a neighbouring anion. The anion's cloud is **distorted** towards the cation.
- Some electron density now sits **between** the two nuclei, so the bond is partly **shared**: partial covalent character.
- This extra attraction makes the real (experimental) lattice energy **more exothermic** than the purely ionic model predicts.



purely ionic: both ions spherical

polarised: electron density pulled towards the cation (covalent character

A small, highly charged cation distorts the electron cloud of a large anion, so some electron density is shared between the ions.

Which ions do it most

	Increases with…	Examples (most → least)
Polarising power of the cation	**high charge density**: high charge, small radius	$Al^{3+} > Mg^{2+} > Na^{+}$; $Li^{+} > Na^{+} > K^{+}$
Polarisability of the anion	**large radius** (electrons far from the nucleus, well shielded) and high charge	$I^{-} > Br^{-} > Cl^{-} > F^{-}$; $S^{2-} > O^{2-}$

The most covalent ionic compounds pair a small, highly charged cation with a large anion: AlI_3, MgI_2, LiI. The least covalent pair a large, low-charge cation with a small anion: CsF, NaF.

WORKED EXAMPLE

Worked example: ranking covalent character

Which has more covalent character, NaF or $MgCl_2$?

Mg^{2+} has a higher charge and smaller radius than Na^{+}, so a **higher charge density** and more **polarising power**.

Cl^{-} is larger than F^{-}, so it is **more polarisable**.

Both effects point the same way: **$MgCl_2$** has more covalent character, and its experimental lattice energy will be further from (more exothermic than) the theoretical value.

MARK-SCHEME WORDING

- "the calcium ion is polarising" and "the iodide ion … is polarised / distorted" (WCH14 Jan 2020 Q19).
- "the magnesium ion is more polarising than the sodium ion"; "the chloride ion is larger / more polarisable than the fluoride ion" (WCH14 Jan 2024 Q17).

Often asked as: inside the 12.14 questions.

COMMON MISTAKE

Saying "the chlorine is polarised". It's the **chloride ion**.

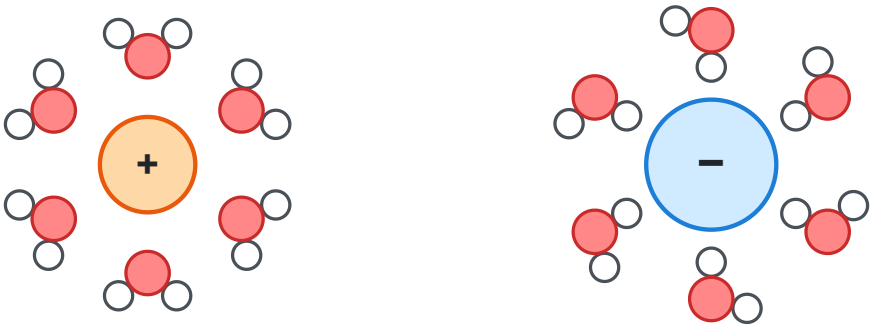
SPEC 12.16 Definitions: enthalpy change of solution and of hydration

WHAT THIS SPEC POINT ASKS

Define the enthalpy change of solution, $\Delta_{sol}H$, and the enthalpy change of hydration of an ion, $\Delta_{hyd}H$.

Term	Definition	Example	Sign
Enthalpy change of solution, $\Delta_{sol}H$	enthalpy change when **one mole of an ionic solid dissolves** in enough water to form an **infinitely dilute solution**	$NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$	can be positive or negative
Enthalpy change of hydration, $\Delta_{hyd}H$	enthalpy change when **one mole of gaseous ions** dissolves in enough water to form an **infinitely dilute solution**	$Na^+(g) + aq \rightarrow Na^+(aq)$ $Cl^-(g) + aq \rightarrow Cl^-(aq)$	always **negative**

- **Infinitely dilute**: so much water that adding more gives **no further heat change**. Then the ions don't interact with each other, only with water.
- Hydration is exothermic because of **ion–dipole attractions**: the δ− oxygen of water points towards a cation; the δ+ hydrogens point towards an anion. New attractions form; no bonds break.



δ− oxygen atoms face the cation

δ+ hydrogen atoms face the anion

Hydrated ions. Oxygen atoms are red and hydrogen atoms white.

MARK-SCHEME WORDING

- Hydration: "enthalpy change when 1 mol of gaseous ions" is "completely hydrated / forms an infinitely dilute solution / dissolves until there is no further heat change" (WCH14 Jun 2025 Q19). DO NOT AWARD "heat needed" or "dissolved in 1 mol of water". IGNORE "standard conditions".
- Hydration enthalpy of an ion is always negative (WCH14 Oct 2022 Q4).

Often asked as: define $\Delta_{hyd}H$; MCQ on which equation shows hydration or solution.

COMMON MISTAKE

Defining hydration for the solid compound, or for 1 mol of water. It is for **gaseous ions**.

SPEC 12.17 Energy cycles for the enthalpy change of solution

WHAT THIS SPEC POINT ASKS

Use energy cycles and energy-level diagrams to calculate the enthalpy change of solution of an ionic compound from hydration enthalpies and the lattice energy.

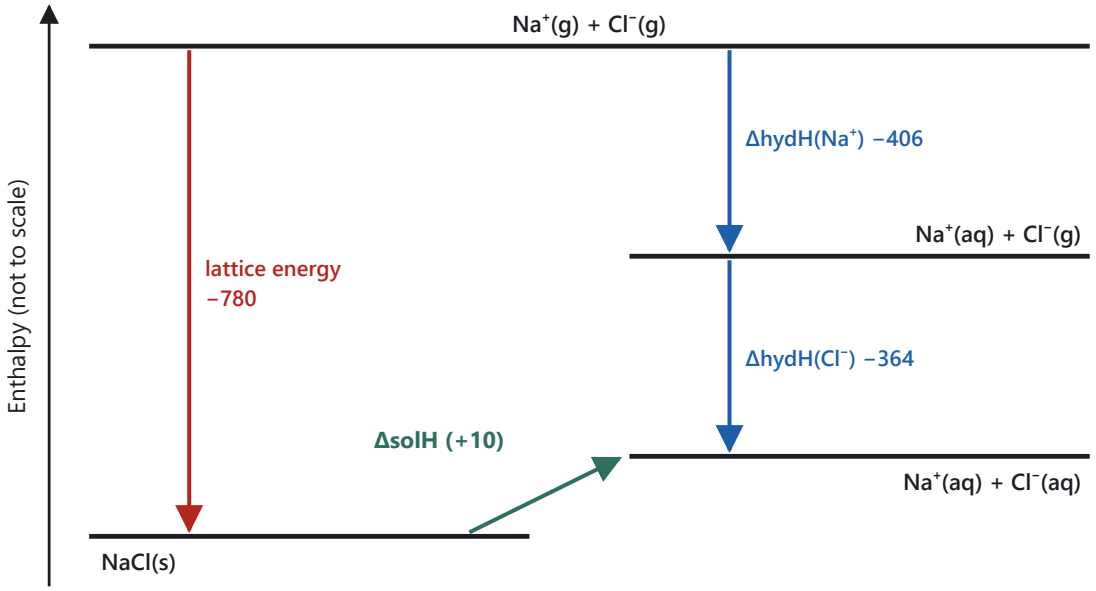
The cycle

Two routes from the solid to the solution:

- Direct: $\mathbf{\Delta_{sol}H}$.
- Indirect: break the lattice into gaseous ions ($\mathbf{-LE}$, endothermic, because LE is defined as the exothermic formation), then hydrate each ion ($\mathbf{\Sigma\Delta_{hyd}H}$).

$\Delta_{sol}H = \Sigma\Delta_{hyd}H - LE$

For $MgCl_2$ this is $\Delta_{sol}H = \Delta_{hyd}H(Mg^{2+}) + \mathbf{2} \times \Delta_{hyd}H(Cl^-) - LE(MgCl_2)$.



$\Delta solH = \Delta hydH(Na^+) + \Delta hydH(Cl^-) - LE = -770 - (-780) = +10$ kJ mol^{-1}

NaCl: the lattice energy and the total hydration enthalpy are both large and nearly cancel, leaving a small $\Delta_{sol}H$.

Enthalpy change of solution

$$\Delta_{\text{sol}}H = -\text{LE} + \sum \Delta_{\text{hyd}}H$$

LE: lattice energy (negative, for formation) $\sum \Delta_{\text{hyd}}H$: hydration enthalpies of all the ions

WORKED EXAMPLE

Worked example 1: $\Delta_{sol}H$ of NaCl

LE(NaCl) = −780; $\Delta_{hyd}H(Na^+)$ = −406; $\Delta_{hyd}H(Cl^-)$ = −364 kJ mol^{-1}

$\Delta_{sol}H$ = (−406 + −364) − (−780) = −770 + 780 = **+10 kJ mol^{-1}** (slightly endothermic)

WORKED EXAMPLE

Worked example 2: a missing hydration enthalpy

For $MgCl_2$: $\Delta_{sol}H = -155$; LE = −2526; $\Delta_{hyd}H(Cl^-) = -364$. Find $\Delta_{hyd}H(Mg^{2+})$.

$\Delta_{hyd}H(Mg^{2+}) = \Delta_{sol}H + LE - 2 \times \Delta_{hyd}H(Cl^-)$

$= -155 + (-2526) - 2(-364) = -155 - 2526 + 728 =$ **−1953 kJ mol⁻¹**

WORKED EXAMPLE

Worked example 3: from $\Delta_{sol}H$ to a temperature change

2.00 g of anhydrous $MgCl_2$ (M_r 95.3) dissolves in 50.0 g of water. $\Delta_{sol}H = -155$ kJ mol⁻¹; c = 4.18 J g⁻¹ K⁻¹.

Moles = 2.00 ÷ 95.3 = 0.02099 mol

Heat released = 155 000 × 0.02099 = 3253 J

ΔT = 3253 ÷ (50.0 × 4.18) = **+15.6 °C** (temperature rises because $\Delta_{sol}H$ is negative)

MARK-SCHEME WORDING

- Expression: "$\Delta_{sol}H = \Delta_{hyd}H(M^{n+}) + \Delta_{hyd}H(X^{n-})$ − lattice energy"; signs reversed scores nothing (WCH14 Jun 2022 Q16).
- Cycle: "enthalpy of solution arrow up from the solid" (when endothermic); "lattice energy … gaseous ions, arrow down"; "hydration arrows down to both aqueous ions" (WCH14 Jan 2025 Q19).
- $CuCl_2$: $\Sigma\Delta_{hyd}H = -2100 + 2(-378) = -2856$; $\Delta_{sol}H = -2856 - (-2811) =$ **−45 kJ mol⁻¹**; +45 scores only 1 (WCH14 Jun 2025 Q19).

Often asked as: complete a solution cycle; calculate $\Delta_{sol}H$ or a missing hydration enthalpy; link to a calorimetry experiment.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Calorimetry of dissolving** (q = mcΔT with the mass of **water**, moles from the correct formula mass, sign from the temperature change): WCH14 Oct 2024 Q20, Jun 2025 Q19.
- Hydrated salts: use the M_r of the hydrate (e.g. $CuSO_4 \cdot 5H_2O$) when that's what was weighed (WCH14 Jun 2025 Q19).

COMMON MISTAKE

Not doubling $\Delta_{hyd}H(Cl^-)$ for $MgCl_2$ or $CuCl_2$. Getting the sign of LE wrong in the expression.

In calorimetry, using the mass of the salt (or salt + water) as m.

SPEC 12.18 How ionic charge and radius affect lattice and hydration enthalpies

WHAT THIS SPEC POINT ASKS

Understand the effect of ionic charge and ionic radius on the hydration enthalpies of ions and the lattice energy of ionic compounds.

The rule

Both lattice energy and hydration enthalpy come from **electrostatic attraction**, so both become **more exothermic** when:

- the ions have **higher charges** (stronger attraction), and
- the ions are **smaller** (charges closer together: closer to each other in the lattice, or closer to the water dipoles).

In one phrase: higher **charge density** → more exothermic.

Lattice energy

Compare	Lattice energy	Reason
NaCl v KCl v CsCl	about −780, −711, −661 kJ mol^{-1}	cation larger → ions further apart → weaker attraction → less exothermic
NaF v NaCl v NaI	about −918, −780, −705	anion larger → less exothermic
NaCl v MgO	about −780 v −3800	2+ and 2− charges (and smaller ions) → much more exothermic

Hydration enthalpy

Cation	$\Delta_{hyd}H$ / kJ mol^{-1}	Cation	$\Delta_{hyd}H$	Anion	$\Delta_{hyd}H$
Li^+	−519	Mg^{2+}	−1920	F^-	−506

Cation	$\Delta_{hyd}H$ / kJ mol^{-1}	Cation	$\Delta_{hyd}H$	Anion	$\Delta_{hyd}H$
Na^+	−406	Ca^{2+}	−1650	Cl^-	−364
K^+	−322	Sr^{2+}	−1480	Br^-	−335
Cs^+	−276	Ba^{2+}	−1360	I^-	−293

- Down a group the ions get bigger → hydration less exothermic.
- 2+ ions are hydrated about four times more strongly than 1+ ions of similar size.

EXAM TIP

Compare like with like: change **one** thing (charge **or** radius) and keep the other ion the same. Then say *why*: "smaller ion, so its charge is closer to the δ− oxygen of water, so stronger ion–dipole attraction, so more energy released".

MARK-SCHEME WORDING

- Lattice energy "becomes less exothermic as the radius of the metal ion increases (and the charge remains the same)"; "weaker forces of attraction between ions" (WCH14 Oct 2020 Q16).
- Cu^+ v Cu^{2+}: "Cu^+ less highly charged … weaker (electrostatic) forces of attraction with water (so less energy is released when hydrated)" (WCH14 Jun 2025 Q19). DO NOT AWARD atoms or "intermolecular forces".
- Ammonium halides: "the bromide ion is larger than the chloride ion"; its hydration enthalpy is "less exothermic"; the lattice energy of ammonium chloride is "more exothermic"; then link both to $\Delta_{sol}H$ (WCH14 Jun 2022 Q16).

Often asked as: MCQ on charge density; explain a trend in lattice energy or hydration enthalpy; explain the difference in $\Delta_{sol}H$ between two salts.

COMMON MISTAKE

Atomic radius instead of **ionic** radius; "halogens" instead of "halides".

"Weaker" or "larger" without saying what is weaker or larger.

SPEC 12.19 Predicting solubility from entropy and enthalpy

WHAT THIS SPEC POINT ASKS

Use entropy changes and enthalpy changes of solution to predict whether an ionic compound dissolves, and discuss the solubility trends from Unit 2.

Is dissolving feasible?

A salt dissolves if **ΔS_{total} for dissolving is positive**:

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings} = \Delta S_{system} - \Delta_{sol}H/T$$

- **Exothermic $\Delta_{sol}H$**: $\Delta S_{surroundings}$ is positive, so dissolving is usually feasible.
- **Small positive $\Delta_{sol}H$** (NaCl, KCl, NH_4NO_3): feasible as long as **$\Delta S_{system} > \Delta_{sol}H/T$**.
- **Large positive $\Delta_{sol}H$**: $\Delta S_{surroundings}$ is very negative, so the salt is insoluble (or only sparingly soluble).
- ΔS_{system} itself is the balance between the lattice breaking up (+) and water being ordered round the ions (−) (12.5).

WORKED EXAMPLE

Worked example: how positive must ΔS_{system} be?

A salt has $\Delta_{sol}H$ = +25.7 kJ mol^{-1}. It dissolves at 298 K. What can you say about ΔS_{system}?

$\Delta S_{surroundings}$ = −25 700 ÷ 298 = −86.2 J K^{-1} mol^{-1}

For $\Delta S_{total} > 0$, **ΔS_{system} must be more positive than +86.2 J K^{-1} mol^{-1}**. (For NH_4NO_3 it is +108.7, so it dissolves.)

Unit 2 trends: Group 2 hydroxides and sulfates

Both lattice energy and hydration enthalpy get **less exothermic** as the cation gets bigger down Group 2. What matters is **which falls faster**, because $\Delta_{sol}H = \Sigma\Delta_{hyd}H - LE$.

	Group 2 **sulfates** (large anion)	Group 2 **hydroxides** (small anion)
Lattice energy down the group	changes **only a little**: the inter-ion distance is dominated by the big SO_4^{2-}	becomes much less exothermic: OH^- is small, so the cation's size matters a lot

	Group 2 **sulfates** (large anion)	Group 2 **hydroxides** (small anion)
Hydration enthalpy of M^{2+}	becomes **much** less exothermic	becomes less exothermic, but by less than the lattice energy
$\Delta_{sol}H$ down the group	becomes **less exothermic / more endothermic**	becomes **more exothermic / less endothermic**
Solubility down the group	**decreases** ($BaSO_4$ is insoluble)	**increases** ($Mg(OH)_2$ sparingly soluble, $Ba(OH)_2$ fairly soluble)

Entropy changes play a part as well, so this enthalpy argument is a simplification, but it is the one you are expected to give.

MARK-SCHEME WORDING

- ΔS_{system} for a salt that dissolves: "ΔS_{system} must be positive"; "ΔS_{system} > ($\Delta H/T$ value)"; "ΔS_{total} > 0 as it dissolves" (WCH14 Jun 2021 Q15).
- Least exothermic lattice energy in a 100% ionic model: the compound with **singly charged, larger ions** (WCH14 Jan 2026 Q7).

Often asked as: decide whether a salt dissolves from $\Delta_{sol}H$ and ΔS_{system}; MCQ on lattice energies; link hydration and lattice changes to $\Delta_{sol}H$.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- NaCl dissolving with $\Delta_{sol}H$ = +1.0 kJ mol^{-1} is feasible, so its ΔS_{system} must be positive. The correct MCQ answer is "the change has a positive entropy change of the system", not "a positive $\Delta S_{surroundings}$" (WCH14 Oct 2025 Q4).

COMMON MISTAKE

Using $\Delta_{sol}H$ alone to decide solubility: the entropy terms decide.

Saying solubility trends come only from lattice energy. It's the **balance** of lattice energy against hydration.

Quick checks

QUICK CHECK

Which equation represents the lattice energy of magnesium oxide?

- $Mg(s) + \frac{1}{2}O_2(g) \rightarrow MgO(s)$
- $MgO(s) \rightarrow Mg^{2+}(g) + O^{2-}(g)$
- ✓ $Mg^{2+}(g) + O^{2-}(g) \rightarrow MgO(s)$
- $Mg^{2+}(aq) + O^{2-}(aq) \rightarrow MgO(s)$

Lattice energy is the formation of 1 mol of solid from its gaseous ions. The first option is the enthalpy of formation; the second is the lattice breaking up (same size, opposite sign); aqueous ions are wrong because hydration energy would be involved.

QUICK CHECK

Use the data to find the first electron affinity of bromine. Data / kJ mol^{-1}: $\Delta_f H$(KBr) = −394; $\Delta_{at} H$(K) = +89; first IE(K) = +419; $\Delta_{at} H$(Br) = +112; LE(KBr) = −689.

- +325 kJ mol^{-1}
- −1703 kJ mol^{-1}
- −915 kJ mol^{-1}
- ✓ −325 kJ mol^{-1}

$\Delta_f H = \Delta_{at} H(K) + IE + \Delta_{at} H(Br) + EA + LE$, so EA = −394 − (89 + 419 + 112 − 689) = −394 − (−69) = −325 kJ mol^{-1}. A positive answer comes from a sign error with the lattice energy.

QUICK CHECK

The Born–Haber lattice energy of a compound is much more exothermic than its theoretical lattice energy. What does this show?

- The ions are larger than expected.
- The bonding has significant covalent character.
- The compound is purely ionic.
- The compound will be very soluble in water.

The theoretical value assumes perfectly spherical ions. A polarising cation distorts the anion, putting electron density between the ions; this covalent character strengthens the lattice beyond what the ionic model predicts.

QUICK CHECK

Which change has the largest positive ΔS_{system}?

- $H_2O(g) \rightarrow H_2O(l)$
- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
- $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
- $Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$

Only the decomposition of calcium carbonate creates a gas from a solid. The other three all reduce the amount or freedom of gas particles, so their ΔS_{system} is negative.

QUICK CHECK

A reaction has $\Delta H = -120$ kJ mol^{-1} and $\Delta S_{system} = -150$ J K^{-1} mol^{-1}. Above what temperature is it no longer feasible?

- 0.8 K
- 800 K
- 1250 K
- It is feasible at all temperatures.

Feasible while $\Delta G = \Delta H - T\Delta S$ is negative. $\Delta G = 0$ when $T = \Delta H/\Delta S = -120\,000 / -150 = 800$ K. Above that, $-T\Delta S$ (positive) outweighs the negative ΔH. Forgetting to convert kJ to J gives 0.8 K.

QUICK CHECK

Why is the hydration enthalpy of Mg^{2+} much more exothermic than that of Na^{+}?

- Mg^{2+} has more electrons than Na^{+}.
- Mg^{2+} forms covalent bonds with water.
- Mg^{2+} is smaller and has a larger charge, so it attracts the δ− oxygen of water molecules more strongly.
- Mg^{2+} is surrounded by hydrogen atoms of water molecules.

Hydration energy comes from ion–dipole attraction. Mg^{2+} has a much higher charge density than Na^{+} (twice the charge, smaller radius), so water's δ− oxygen atoms are pulled in closer and more strongly. Cations are surrounded by oxygen, not hydrogen, atoms.

QUICK CHECK

The conversion of diamond into graphite has a negative ΔG at room temperature, yet diamonds do not change. Why?

- ΔS_{total} for the change is negative.
- The activation energy is so high that the rate is negligible.
- The equilibrium constant is less than 1.
- Graphite has a lower entropy than diamond.

A negative ΔG means ΔS_{total} is positive and K > 1, so the change is feasible. Feasibility says nothing about rate: diamond is kinetically stable because the activation energy for rearranging its strong covalent network is enormous.